

WHITE PAPER 2026

The operational intelligence framework

How Leading Organisations Create Better Operational Understanding

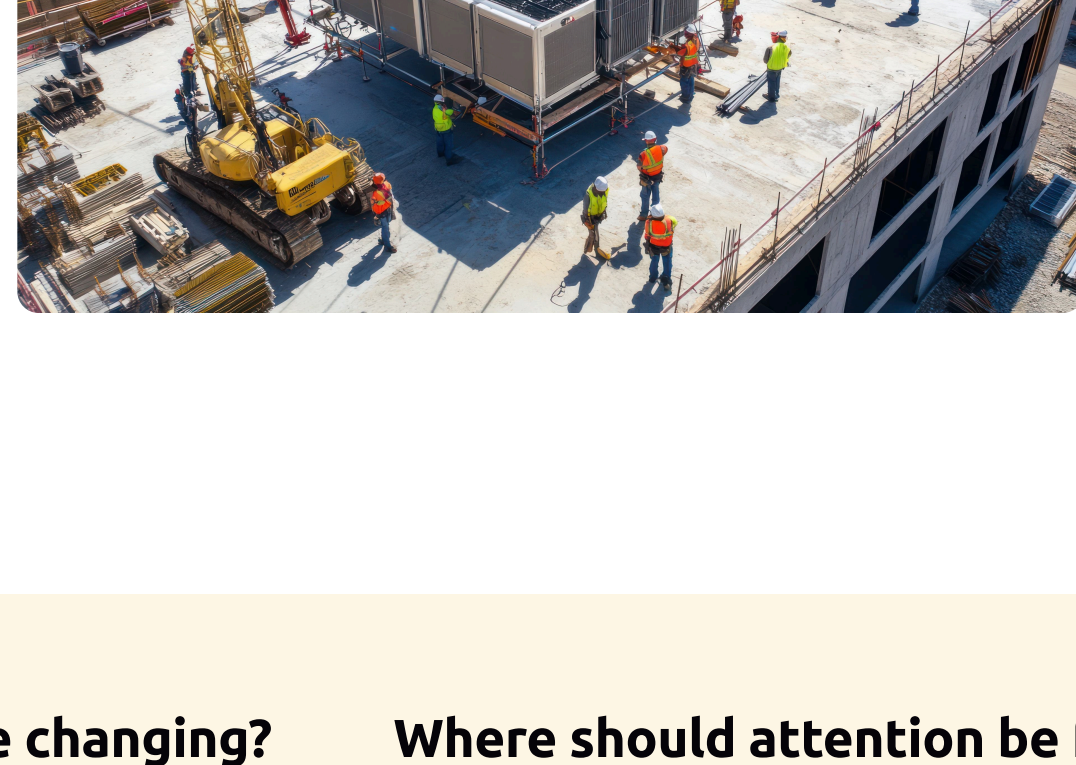
Operational Intelligence for Underground Ventilation

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Executive summary

Operations are becoming increasingly connected, data-rich and complex. Workforce systems, fleet technologies, environmental monitoring, production platforms, communications systems and operational technologies generate enormous volumes of information every day.

Yet despite significant investment in technology, many organisations still struggle to answer fundamental operational questions:



What is happening? Why is it happening? What conditions are changing? Where should attention be focused?



- The challenge is not a lack of information.
- The challenge is understanding.

Operational Intelligence provides a framework for transforming operational information into actionable insight by helping organisations understand how operational conditions interact across the entire operation.

This paper explores the Operational Intelligence Framework and how organisations can improve safety, productivity and operational performance through better operational understanding.

CHAPTER 1

THE OPERATIONAL COMPLEXITY CHALLENGE

Modern operations have become significantly more complex. Mining operations involve:

 Workfor ces	 Fleet operations	 Equipment	 Environme ntal moni toring
 Production Systems	 Safety controls	 Contract ors	 Infrastruct ure

Tunnelling projects introduce:

 Multiple subcontractors	 Mobile equipment	 Environmental conditions	 Workfronts	 Changing project conditions
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Critical infrastructure environments add:

 Operational continuity requirements	 Security systems	 Distributed assets	 Multiple stakeholder groups
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As complexity increases, understanding becomes more difficult. Most organisations have visibility into individual systems. Few have visibility into how those systems interact.

CHAPTER 2

THE LIMITATIONS OF TRADITIONAL VISIBILITY

Historically, operational technologies have focused on solving specific problems.

Personnel tracking.
Fleet management.
Environmental monitoring.
Production reporting.

Asset tracking.
Each delivers value.
However, each typically operates independently.

This creates fragmented visibility.
Operational teams are forced to review multiple systems to understand what is happening across the operation.

The result is often:
Information overload.
Delayed decision-making.
Operational blind spots.

Missed relationships between operational conditions.
Visibility improves.
Understanding remains limited.

CHAPTER 3

WHAT IS OPERATIONAL INTELLIGENCE?

Operational Intelligence is the ability to understand how operational conditions interact across an environment in real time.
Rather than focusing on individual systems, Operational Intelligence focuses on relationships.

People.
Vehicles.
Equipment.
Environmental conditions.
Production activities.
Operational constraints.

The goal is to create a shared operational understanding that supports better decision-making.
Operational Intelligence helps organisations answer:

What is happening?
Why is it happening?
What factors are contributing?

Who or what is affected?
What requires attention?

CHAPTER 4

THE OPERATIONAL INTELLIGENCE FRAMEWORK

The framework consists of four interconnected layers.

LAYER 1	LAYER 2	LAYER 3	LAYER 4
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Operational Data
Fleet systems
Environmental monitoring
Production systems

Operational technologies
Communications systems
Infrastructure systems

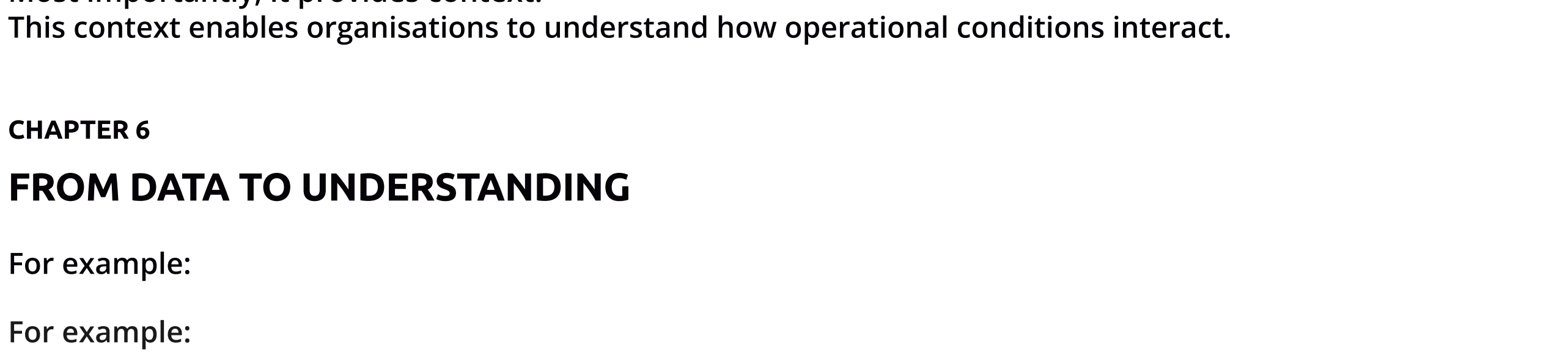
Operational Context
Understanding how operational information relates across the environment.

Relationships.
Dependencies.
Interactions.
Operational conditions.

Operational Insight
Identifying:
Emerging conditions
Developing risks
Performance impacts

Operational constraints
Improvement opportunities

Operational Action
Supporting:
Decision-making
Intervention
Coordination
Planning
Continuous improvement

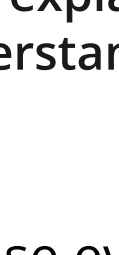
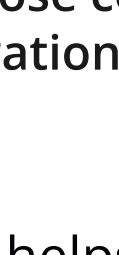


CHAPTER 5

THE OPERATIONAL DIGITAL TWIN

The Operational Digital Twin forms the operational foundation of the framework. It creates a live operational representation of the environment by bringing together information from across the operation into a shared operational view.

The Operational Digital Twin provides:

 Operatio nal visibility	 Situatio nal aware ness	 Workfo rce visibility	 Fleet visibility
 Asset visibility	 Environmen tal visibility	 Production awareness	

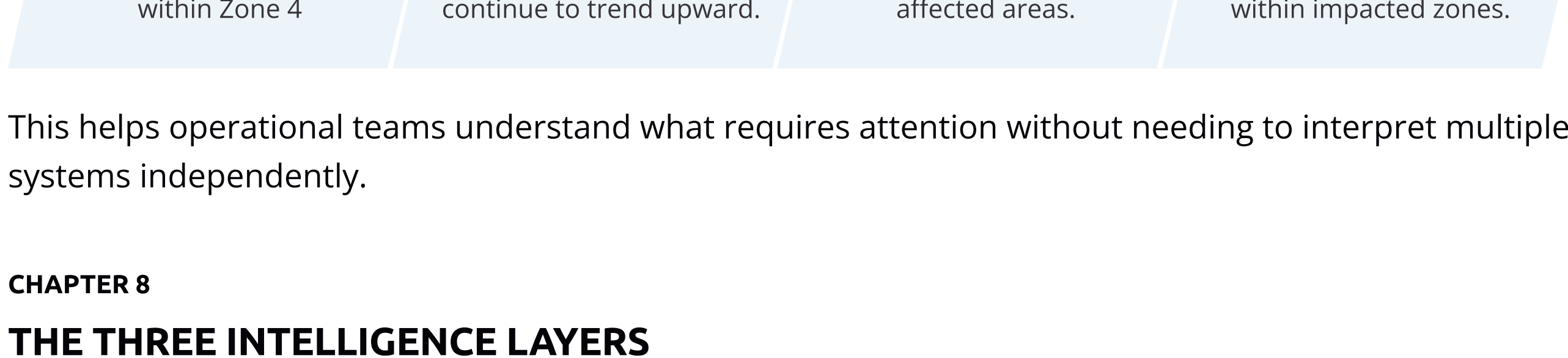
Most importantly, it provides context. This context enables organisations to understand how operational conditions interact.

CHAPTER 6

FROM DATA TO UNDERSTANDING

For example:

For example:



Rather than presenting each condition separately, the platform can explain how those conditions are related and why they matter. This helps organisations move from monitoring information to understanding operational impact.

Traditional systems may present each event separately.
Operational Intelligence identifies the relationship between those events and helps explain why they matter.
This is where understanding begins.

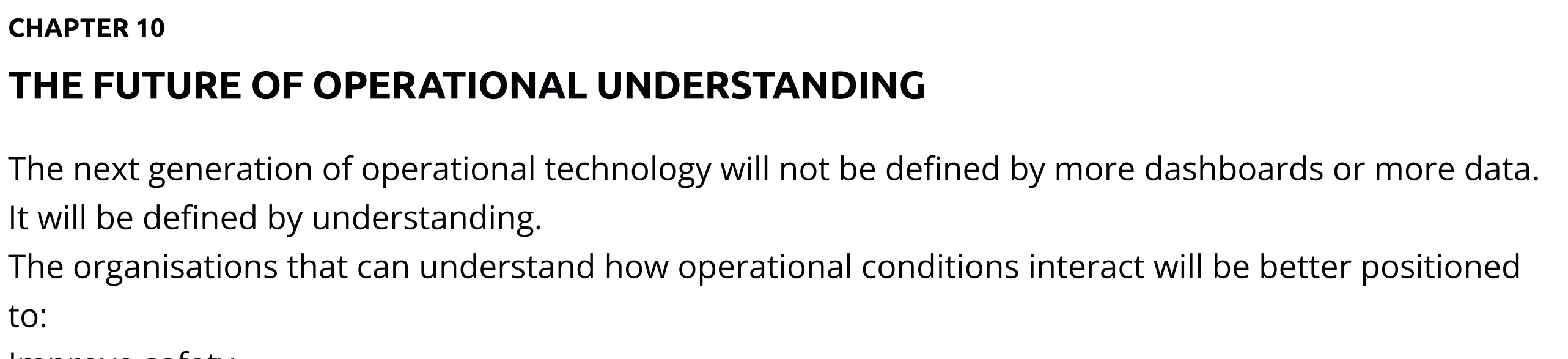
CHAPTER 7

THE ROLE OF AI IN OPERATIONAL INTELLIGENCE

As operations become increasingly complex, AI plays an important role in helping organisations interpret information.
AI should not replace operational expertise.
It should augment it.

By analysing relationships between operational conditions, AI can help transform operational information into simple, actionable narratives.

For example:



This helps operational teams understand what requires attention without needing to interpret multiple systems independently.

CHAPTER 8

THE THREE INTELLIGENCE LAYERS

Operational Intelligence is typically delivered through three complementary intelligence layers.
Operational Digital Twin
Understanding What Is Happening
Fatal Risk Intelligence
Understanding Exposure Before Incidents Occur
Production Intelligence
Understanding The Factors Influencing Operational Performance

Together these layers create a more complete understanding of the operation.

CHAPTER 9

OPERATIONAL INTELLIGENCE ACROSS THE ORGANISATION

Operational Teams
Situational awareness.
Supervisors
Operational coordination.
Management
Operational planning.
Executive Leadership
Portfolio visibility.
Asset Owners
Governance and assurance.
Operational Intelligence creates value throughout the operational hierarchy.

CHAPTER 10

THE FUTURE OF OPERATIONAL UNDERSTANDING

The next generation of operational technology will not be defined by more dashboards or more data. It will be defined by understanding.
The organisations that can understand how operational conditions interact will be better positioned to:

Improve safety.
Increase productivity.
Reduce operational disruption.
Respond faster to change.
Support better decisions.
Operational Intelligence provides the framework for achieving that understanding.

CONCLUSION

Every operation generates information. Very few generate understanding. Operational Intelligence helps organisations bridge this gap by transforming operational complexity into actionable insight. Because better understanding creates better decisions. And better decisions create better outcomes.